

Machine Learning

Unit-II (C): Evaluation Matrices

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How to Evaluate Logistic Regression Model?

- Evaluating the logistic regression model helps assess its performance and ensure it generalizes well to new, unseen data.
- The following metrics are commonly used:

Confusion Matrix

- Confusion matrix is a simple table used to measure how well a classification model is performing.
- It compares the predictions made by the model with the actual results and shows where the model was right or wrong.
- This helps you understand where the model is making mistakes so you can improve it.
- It breaks down the predictions into four categories.

Confusion Matrix

1. **True Positive (TP):** The model correctly predicted a positive outcome i.e., the actual outcome was positive.
2. **True Negative (TN):** The model correctly predicted a negative outcome i.e., the actual outcome was negative.
3. **False Positive (FP):** The model incorrectly predicted a positive outcome i.e., the actual outcome was negative. It is also known as a Type I error.
4. **False Negative (FN):** The model incorrectly predicted a negative outcome i.e., the actual outcome was positive. It is also known as a Type II error.

Confusion Matrix

- It also helps calculate key measures like accuracy, precision and recall which give a better idea of performance especially when the data is imbalanced.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Accuracy

- Accuracy is a fundamental metric used for evaluating the performance of a classification model.
- It tells us the proportion of correct predictions made by the model out of all predictions.

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total}}$$

Accuracy

- While accuracy provides a quick snapshot, it can be misleading in cases of imbalanced datasets.
- For example, in a dataset with 90% class A and 10% class B, a model predicting only class A will still achieve 90% accuracy but it will fail to identify any class B instances.
- Accuracy is good but it gives a False Positive sense of achieving high accuracy.
- The problem arises due to the possibility of misclassification of minor class samples being very high.

Precision

- It measures how many of the positive predictions made by the model are actually correct.
- It's useful when the cost of false positives is high such as in medical diagnoses where predicting a disease when it's not present can have serious consequences.
- Precision helps ensure that when the model predicts a positive outcome, it's likely to be correct.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Recall

- Recall or Sensitivity measures how many of the actual positive cases were correctly identified by the model.
- It is important when missing a positive case (false negative) is more costly than false positives.
- In scenarios where catching all positive cases is important (like disease detection), recall is a key metric.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

F1 Score

- The F1 Score is the **harmonic mean of precision and recall**.
- It is **useful when we need a balance between precision and recall** as it combines both into a single number.
- A **high F1 score means the model performs well on both metrics**. Its range is [0,1].
- Lower recall and higher precision gives us great accuracy but then it misses a large number of instances.
- More the F1 score better will be performance.
- It can be expressed mathematically in this way:

$$\text{F1 Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Logarithmic Loss (Log Loss)

- Log Loss measures the uncertainty of the model's predictions.
- It is calculated by penalizing the model for assigning low probabilities to the correct classes.
- This metric is used in multi-class classification and is helpful when we want to assess a model's confidence in its predictions.
- If there are N samples belonging to the M class, then we calculate the average Log loss in this way:

$$\text{Log Loss} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} \cdot \log(p_{ij})$$

Where:

- y_{ij} = Actual class (0 or 1) for sample i and class j
- p_{ij} = Predicted probability for sample i and class j
- The goal is to minimize Log Loss, as a lower Log Loss shows higher prediction accuracy.

Area Under Curve (AUC)

- It is useful for binary classification tasks.
- The AUC value represents the probability that the model will rank a randomly chosen positive example higher than a randomly chosen negative example.
- AUC ranges from 0 to 1 with higher values showing better model performance.

True Positive Rate (TPR)

- Also known as sensitivity or recall, the True Positive Rate measures how many actual positive instances were correctly identified by the model.
- It answers the question: "Out of all the actual positive cases, how many did the model correctly identify?"

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

True Negative Rate (TNR)

- Also called *specificity*, the True Negative Rate measures how many actual negative instances were correctly identified by the model.
- It answers the question: "Out of all the actual negative cases, how many did the model correctly identify as negative?"

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

False Positive Rate (FPR)

- It measures how many actual negative instances were incorrectly classified as positive.
- It's a key metric when the cost of false positives is high such as in fraud detection.

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

False Negative Rate (FNR)

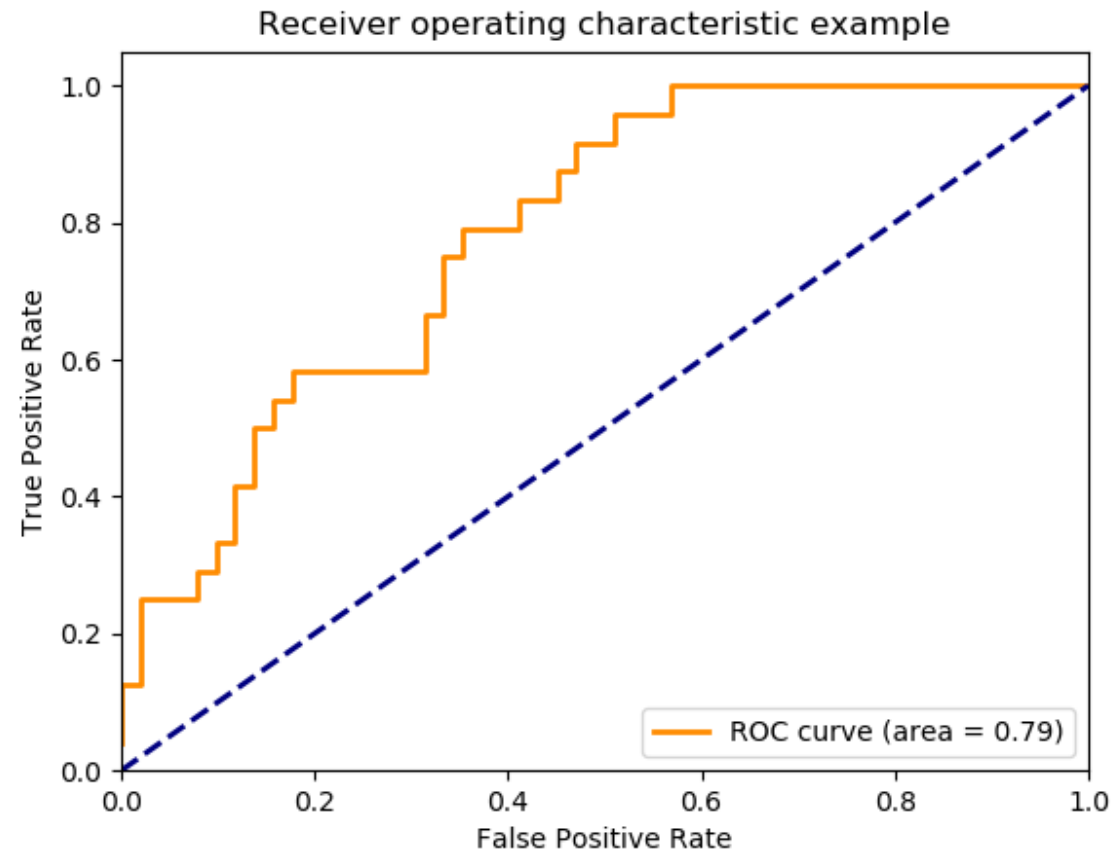
- It measures how many actual positive instances were incorrectly classified as negative.
- It answers: "Out of all the actual positive cases, how many were misclassified as negative?"

$$\text{FNR} = \frac{\text{FN}}{\text{FN} + \text{TP}}$$

ROC Curve

- It is a graphical representation of the True Positive Rate (TPR) vs the False Positive Rate (FPR) at different classification thresholds.
- The curve helps us visualize the trade-offs between sensitivity (TPR) and specificity (1-FPR) across various thresholds.
- Area Under Curve (AUC) quantifies the overall ability of the model to distinguish between positive and negative classes.
 - $AUC = 1$: Perfect model (always correctly classifies positives and negatives).
 - $AUC = 0.5$: Model performs no better than random guessing.
 - $AUC < 0.5$: Model performs worse than random guessing (showing that the model is inverted).

ROC Curve



Question 1

A binary classifier outputs probability scores for 10 samples:

Sample	True Label	Predicted Probability
1	1	0.95
2	1	0.90
3	1	0.80
4	1	0.60
5	1	0.40
6	0	0.70
7	0	0.50
8	0	0.30
9	0	0.20
10	0	0.10

Question 1

1. Using threshold = 0.5, Compute:
 - A. TP, FP, TN, FN
 - B. TPR (Recall)
 - C. FPR
2. Using threshold = 0.7, Compute:
 - A. TP, FP, TN, FN
 - B. TPR
 - C. FPR
3. Plot the two ROC points: FPR and TPR.

Solution 1

- Predicted Positive (≥ 0.5)
 - TP = 4 (1, 2, 3, 4)
 - FP = 2 (6, 7)
 - FN = 1 (sample 5 $\rightarrow$ 0.40)
 - TN = 3 (8, 9, 10)
- $TPR = \frac{TP}{TP+FN} = \frac{4}{5} = 0.8$
- $FPR = \frac{FP}{FP+TN} = \frac{2}{5} = 0.4$

Solution 1

- Predicted Positive (≥ 0.7):
 - TP = 3 (1, 2, 3)
 - FP = 1 (6)
 - FN = 2 (4, 5)
 - TN = 4 (7, 8, 9, 10)
- $TPR = \frac{TP}{TP+FN} = \frac{3}{5} = 0.6$
- $FPR = \frac{FP}{FP+TN} = \frac{1}{5} = 0.2$

Solution 1

ROC uses:

$$(x, y) = (\text{FPR}, \text{TPR})$$

So we get:

Threshold 0.5 $\rightarrow$ (0.4, 0.8)

Threshold 0.7 $\rightarrow$ (0.2, 0.6)

Question 2

A learning model is developed to detect Diabetic Retinopathy (Positive class = 1) from retinal images. The model outputs probability scores for 12 patients as shown below:

Patient	True Label	Predicted Probability
P1	1	0.95
P2	1	0.85
P3	1	0.75
P4	1	0.55
P5	1	0.45
P6	0	0.80
P7	0	0.60
P8	0	0.50
P9	0	0.40
P10	0	0.30
P11	0	0.20
P12	0	0.10

Question 2

1. Task A – Threshold = 0.50
 - A. Construct the Confusion Matrix and compute: Precision, Recall (TPR), FPR, Specificity, Accuracy, and F1-score.
2. Task B – Threshold = 0.70
 - A. Construct the new Confusion Matrix and compute: Precision, Recall and FPR.
3. Task C – ROC Curve
 - A. Write the ROC points (FPR, TPR) for:
 - a) Threshold = 0.70
 - b) Threshold = 0.50
 - c) Threshold = 1.0
 - d) Threshold = 0.0
 - B. Plot these points on ROC space.
 - C. Approximate the AUC using trapezoidal rule.

Solution 2

Predicted Positive (≥ 0.5): P1, P2, P3, P4, P6, P7, P8

Confusion Matrix

	Predicted 1	Predicted 0
Actual 1	TP = 4	FN = 1
Actual 0	FP = 3	TN = 4

Solution 2

- **F1 – score** = $\frac{2PR}{P+R} = \frac{2(0.571)(0.8)}{0.571+0.8} \approx 0.67$
- **Accuracy** = $\frac{TP+TN}{12} = \frac{8}{12} = 0.67$
- **Specificity** = $\frac{TN}{TN+FP} = \frac{4}{7} \approx 0.57$

Solution 2

Predicted Positive (≥ 0.7): P1, P2, P3, P6

Confusion Matrix:

	Predicted 1	Predicted 0
Actual 1	TP = 3	FN = 2
Actual 0	FP = 1	TN = 6

Solution 2

Solution 2

ROC Curve Points

Threshold = 1.0

- No probability ≥ 1.0
- $TP = 0$
- $FP = 0$
- $TPR = 0, FPR = 0$
- ROC point $\rightarrow (0, 0)$

Threshold = 0.70

- $FPR = 0.14, TPR = 0.6$
- ROC point $\rightarrow (0.14, 0.6)$

Solution 2

Threshold = 0.50

- $FPR = 0.43, TPR = 0.8$
- ROC point $\rightarrow (0.43, 0.8)$

Threshold = 0.0

- All predicted positive
- $TP = 5$
- $FP = 7$
- $TPR = 1, FPR = 1$
- ROC point $\rightarrow (1, 1)$

Solution 2

ROC Points in Order

$$(0, 0) \rightarrow (0.14, 0.6) \rightarrow (0.43, 0.8) \rightarrow (1, 1)$$

Solution 2

- The ROC curve is piecewise linear between points.
- Area between two points forms a **trapezoid**.
- Area of trapezoid:

$$\text{Area} = (x_2 - x_1) \frac{(y_1 + y_2)}{2}$$

Where:

Solution 2

Approximate AUC (Trapezoidal Rule)

We compute area piecewise:

Between (0,0) and (0.14,0.6)

$$0.14 \times \frac{0 + 0.6}{2} = 0.042$$

Between (0.14,0.6) and (0.43,0.8)

$$0.29 \times \frac{0.6 + 0.8}{2} = 0.203$$

Between (0.43,0.8) and (1,1)

$$0.57 \times \frac{0.8 + 1}{2} = 0.513$$

Solution 2

Total AUC:

$$AUC \approx 0.042 + 0.203 + 0.513 = 0.758$$

$$AUC \approx 0.76$$

Thank You!